Case Study 5: Warehouse Inventory Reconciliation

1. Problem Statement

For multiple products, compare opening stock plus incoming stock minus dispatched stock with the recorded closing stock. Identify discrepancies and calculate replenishment needs.

2. Objectives

Use lists to store product records.

Calculate expected closing stock.

Identify stock differences and shortages.

3. Assumptions

The program processes three products.

All quantities must be non-negative integers.

Target stock is entered for each product.

Discrepancy = recorded closing stock − expected closing stock.

4. Algorithm

1. Start.

2. For each product, input name, opening stock, incoming stock, dispatched stock, recorded closing stock, and target stock.

3. Calculate expected closing stock.

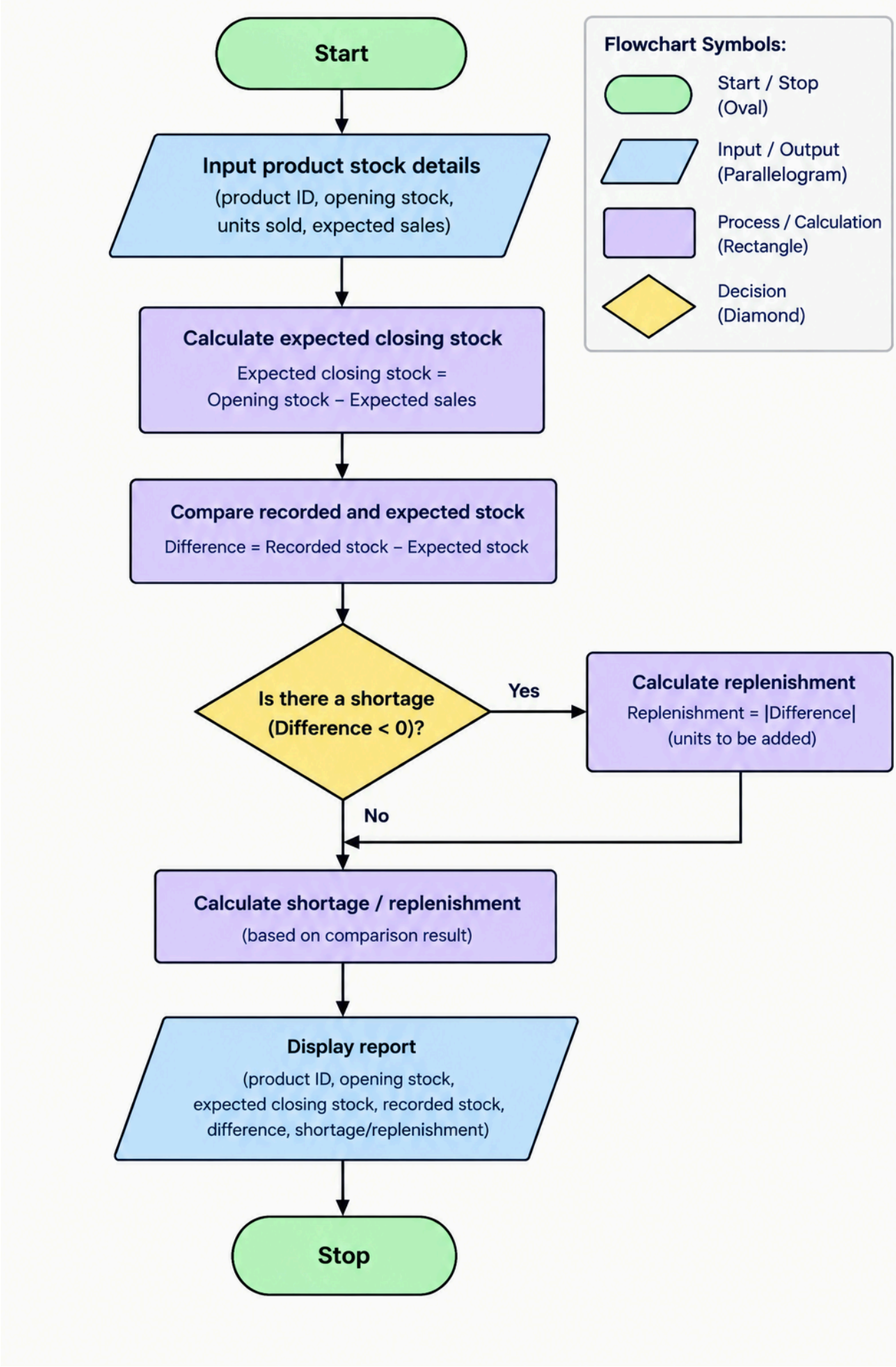
4. Compare expected and recorded closing stock.

5. Calculate replenishment needed to reach target stock.

6. Display each product’s reconciliation report.

7. Stop.

5. Flowchart (text form)



6. Pseudocode

```
START
REPEAT for 3 products: INPUT name, opening, incoming, dispatched, recorded, target
expected = opening + incoming - dispatched
difference = recorded - expected
IF expected < 0: report invalid stock movement
replenishment = maximum of 0 and target - recorded
DISPLAY expected, difference, replenishment
STOP
```

7. Complete Python Source Code

```
products = []

for i in range(3):
```



```
 print(f"
Product {i + 1}")
  name = input("Product name: ")
```

opening = int(input("Opening stock: "))

incoming = int(input("Incoming quantity: "))

dispatched = int(input("Dispatched quantity: "))

closing = int(input("Recorded closing stock: "))

target = int(input("Target stock: "))

```
  if min(opening, incoming, dispatched, closing, target) < 0:
   print("Invalid record: quantities cannot be negative.")
```

continue

```
  expected = opening + incoming - dispatched
   if expected < 0:
   print("Invalid record: dispatched quantity exceeds available stock.")
```

continue

```
  difference = closing - expected
   replenishment = max(0, target - closing)
```

products.append((name, expected, closing, difference, replenishment))

```
print("
----- INVENTORY RECONCILIATION -----")
for name, expected, closing, difference, replenishment in products:
 print("
Product:", name)
 print("Expected closing stock:", expected)
 print("Recorded closing stock:", closing)
 if difference == 0:
 print("Status: No discrepancy")
 else:
 print("Discrepancy (recorded - expected):", difference)
 print("Replenishment required:", replenishment)
```

8. Test Case and Expected Output

One example product: opening 50, incoming 20, dispatched 30, recorded closing 38, target 60.

Expected closing stock: 40 Recorded closing stock: 38 Discrepancy (recorded - expected): -2 Replenishment required: 22

9. Conclusion

The program compares expected and recorded stock, detects discrepancies, and calculates replenishment requirements.